



UITs

Future will be better than the past

University of Information Technology & Sciences

An initiative of **PHP** Family

Microprocessors, Microcontrollers and Assembly Language Lab

Lab Report - 3

Course Title : Microprocessors, Microcontrollers and Assembly
Language Lab

Course Code : CSE0611324

Submitted to:

Teacher's Name	Designation
Tahmina Yeasmin Lima	Lecturer
Abrar Mahbub Tanim	Lecturer

Submitted by:

Student Name : Gaus Saraf Murady

Student ID : 0432410005101088

Semester : Autumn 2026

Batch : 55

Department : CSE

Date of Submission: 30.08.26

Experiment No : 3

Experiment Name: Completing the Observation Table from the Code

Introduction:

In this lab, we operated on Hex values to do AND, OR, Shift Left and Right, Rotate Left and Right operations. Then display the result values as High Nibble and Low Nibble, since AL can only display 4 bits at a time (1 nibble = 4 bits = half byte).

Code:

```
.model small
.stack 100h
.data
    num1 db 018h
    num2 db 02Ah
    num3 db 0C4h
    num4 db 06Bh
    num5 db 0FFh
    num6 db 07Ah

    msg1 db "num1 : $"
    msg2 db "num2 : $"
    msg3 db "num3 : $"
    msg4 db "num4 : $"
    msg5 db "num5 : $"
    msg6 db "num6 : $"

.code
    main proc
        mov ax, @data
        mov ds, ax

        ;
        #####
        #####

        ;NUM 1 | AND op

msg1    mov dx, offset

        mov ah, 09h
        int 21h

        mov al, num1
        and al, 0Fh
        mov bl, al

        ; PRINT

        mov al, bl
        shr al, 4
        call print_hex

        mov al, bl
        and al, 0Fh
        call print_hex

        mov ah, 4Ch
        int 21h

        ; NEWLINE and
CARRY BACK

        mov ah, 2
        mov dl, 10
        int 21h
```

```
        mov ah, 2
        mov dl, 13
        int 21h

        ;
        #####
        #####

        ;NUM 2 | OR op

msg2    mov dx, offset

        mov ah, 09h
        int 21h

        mov al, num2
        or al, 0D1h
        mov bl, al

        ; PRINT

        mov al, bl
        shr al, 4
        call print_hex

        mov al, bl
        and al, 0Fh
        call print_hex

        mov ah, 4Ch
        int 21h

        ; NEWLINE and
CARRY BACK

        mov ah, 2
        mov dl, 10
        int 21h

        mov ah, 2
        mov dl, 13
        int 21h

        ;
        #####
        #####

        ;NUM 3 | SHL op

msg3    mov dx, offset

        mov ah, 09h
        int 21h

        mov al, num3
        shl al, 3
```

```
        mov bl, al

        ; PRINT

        mov al, bl
        shr al, 4
        call print_hex

        mov al, bl
        and al, 0Fh
        call print_hex

        mov ah, 4Ch
        int 21h

        ; NEWLINE and
CARRY BACK

        mov ah, 2
        mov dl, 10
        int 21h

        mov ah, 2
        mov dl, 13
        int 21h

        ;
        #####
        #####

        ;NUM 4 | SHR op

msg4    mov dx, offset

        mov ah, 09h
        int 21h

        mov al, num4
        shr al, 6
        mov bl, al

        ; PRINT

        mov al, bl
        shr al, 4
        call print_hex

        mov al, bl
        and al, 0Fh
        call print_hex

        mov ah, 4Ch
        int 21h

        ; NEWLINE and
CARRY BACK

        mov ah, 2
```

```

mov dl, 10
int 21h

mov ah, 2
mov dl, 13
int 21h

;
#####
#####

;NUM 5 | ROL op

msg5 mov dx, offset

mov ah, 09h
int 21h

mov al, num5
rol al, 2
mov bl, al

; PRINT

mov al, bl
shr al, 4
call print_hex

mov al, bl
and al, 0Fh
call print_hex

mov ah, 4Ch
int 21h

; NEWLINE and
CARRY BACK

```

```

mov ah, 2
mov dl, 10
int 21h

mov ah, 2
mov dl, 13
int 21h

;
#####
#####

;NUM 6 | ROR op

msg6 mov dx, offset

mov ah, 09h
int 21h

mov al, num6
ror al, 7
mov bl, al

; PRINT

mov al, bl
shr al, 4
call print_hex

mov al, bl
and al, 0Fh
call print_hex

mov ah, 4Ch
int 21h

```

```

; NEWLINE and
CARRY BACK

mov ah, 2
mov dl, 10
int 21h

mov ah, 2
mov dl, 13
int 21h

main endp

print_hex proc
add al, 48
cmp al, 57
jbe out
add al, 7

out:
mov dl, al
mov ah, 02h ;
previously set al,
causing unwanted output
int 21h
ret

print_hex endp

end

main

```

Output:

num1: 08

num2: FB

num3: 20

num4: 01

num5: FF

num6: F4

Conclusion:

Num	Instruction	Operation Description	Result(Hex)
18h	AND 0Fh	Used AND operation on AL which stored 018h	08
2Ah	OR D1h	Used OR operation on AL which stored 0D1h	FB
C4h	SHL, 3	Used SHL operation on AL to shift 3 bits left	20
6Bh	SHR, 6	Used SHR operation on AL to shift 6 bits right	01
FFh	ROL, 2	Used RoL operation on AL to rotate 2 bits left	FF
7Ah	ROR, 7	Used SHR operation on AL to rotate 7 bits right	F4